GST EVIDENCE RECONCILIATION

When paperwork agrees, physical reality can disagree.

A GST officer can’t tell which factories are hiding production or inflating tax credits, because every document she checks is declared by the factory itself, while the independent evidence (power use, goods movement, supplier records, bank flows) sits in separate places and is never compared. If this worked, she would start each week with a short list of units whose paperwork contradicts physical reality, each contradiction spelled out, and honest-but-unusual factories cleared.

Decision support for human review

Impact and the current gap

Today, officers must reconcile sources manually, even when physical and financial evidence disagree.

What exists today

Invoice matching finds inconsistencies inside the tax system. It does not reconcile declarations with independent operational evidence.

What this project adds

It puts production, movement, supplier and bank evidence on one factory-month record, with the evidence rows visible to the reviewer.

Why it matters

Prioritise specific contradictions and preserve an explanation for honest operational changes.

Method and the two-signal rule

Four SQL conservation checks run in Exasol Personal. The LLM writes notes only after SQL has selected a case.

01

Material → output

Purchased inputs and power

02

Dispatch → invoice

Goods movement and invoice

03

Stock roll-forward

Opening, production and dispatch

04

Supplier chain

Returns and looping links

PRIORITISE ONLY WHEN TWO INDEPENDENT SOURCE FAMILIES FIRE

A single-source month stays review-only or receives a documented cleared disposition.

Seed C results and limits

Seed A selected the threshold and factory queue rule. Seed C reports the held-out result.

84.01%

Factory-month recall

16.58%

Factory-month false-positive rate

58.67%

Queue precision after K=4 months

917 factories enter the K-month queue. The top 50-case snapshot has 75.00% synthetic precision.

Limits: synthetic data only. Human review remains required. Confidence intervals do not measure generator bias. Power and bank data require lawful access and privacy safeguards.

Exasol Personal and the data-access roadmap

The database runs the evidence views locally. The proposal starts with tax-system data and adds other sources only under lawful access.

20,978,945 raw evidence rows

Ranked workload timing: 48.946 s raw; 6.141 s after the batch ledger refresh. Measured on a MacBook.

Data-access roadmap

1. Tax-system data
Invoices, e-way bills, supplier returns

2. Lawful additional evidence
Power and bank records under authorised access and privacy controls

Germany: the same reconciliation pattern would need local tax rules, lawful data access and independent validation. This deck claims no German deployment or statistic.